JOHAAN TOMY KALLANY

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PhD Applicant — Multiscale Computational Modelling of Particle-Laden Flows

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RESEARCH PROFILE

Graduate of the **Joint MSc in Mathematical Modelling in Engineering (MathMods)**, with research experience at the interface of computational fluid dynamics, particle transport, and numerical modelling of particle-laden flows. My academic training spans numerical analysis and scientific computing, continuum mechanics and PDE modelling, complemented by stochastic modelling and statistical learning. During my research internship at Inria Sophia Antipolis, I developed numerical particle-dynamics solvers and combined Lagrangian particle statistics with Eulerian flow and density diagnostics to investigate how carrier-flow structure, particle inertia, and interparticle collisions shape collective particle dynamics and spatial organisation of particles in controlled carrier flows and near-wall turbulence.

RESEARCH EXPERIENCE

Research Intern — Inria Sophia Antipolis, France

APR. 2026 – SEPT. 2026

Team CALISTO · Supervisors: Dr Christophe Henry and Dr Laetitia Giraldi · Thesis Grade: 17/20

Thesis: ***Collective Dynamics of Colliding Inertial Particles in Prescribed Carrier Flows***

Research Output: First-author manuscript in preparation with Laetitia Giraldi and Christophe Henry.

- **Developed and verified Lagrangian particle solvers** for passive tracers and finite-size inertial particles in Python; combined continuous Stokes-drag dynamics with chronologically resolved hard-particle collisions using operator splitting.
- **Reconstructed and analysed a time-resolved wall-parallel carrier-flow field** from turbulent-channel DNS using validated space–time interpolation; quantified near-wall streak scales from spatial velocity autocorrelations.
- **Quantified preferential sampling and collision-induced particle redistribution** using Eulerian flow diagnostics (speed, strain rate, wall-normal vorticity and in-plane divergence), particle-number fluctuations, DBSCAN cluster morphology and two-point particle-density correlations.
- **Developed pair- and event-conditioned statistical diagnostics** for pair anisotropy, collision-angle selection and same-pair recurrence; quantified uncertainty using bootstrap confidence intervals and sensitivity analyses.
- **Characterised regime-dependent particle organisation in prescribed cellular and sinusoidal-shear flows under chaotic advection**, analysing clustering and dispersion across variations in particle inertia, collision restitution, and stirring amplitude

EDUCATION

Joint MSc in Mathematical Modelling in Engineering: Theory, Numerics, Applications

SEPT. 2024 – SEPT. 2026

Université Côte d'Azur (France) · University of Hamburg (Germany) · University of L'Aquila (Italy) · mathmods.eu

Semester IV — Master's Thesis at Inria Sophia Antipolis

APR. 2026 – SEPT. 2026

Semester III — Université Côte d'Azur, France

SEPT. 2025 – MAR. 2026

Specialisation in **Stochastic Modelling and Statistical Learning**; selected cohort **rankings**: *Stochastic Calculus & Applications* (**2/15**), *Fundamentals of Machine Learning* (**2/20**), and *Statistical Learning from and for Graphs & Geometric Statistics* (**2/10**).

- **Stochastic Calculus, Diffusions & Control:** Brownian motion; Itô calculus and SDEs; Euler–Maruyama discretisation; infinitesimal generators and SDE–PDE correspondence; stochastic control and Hamilton–Jacobi–Bellman equations; mean-field models.
- **Optimisation, Machine Learning & Geometric Statistics:** convex and large-scale optimisation; kernel methods and neural networks; variational inference; stochastic block models and graph learning; geometric statistics on manifolds.

Semester II — University of Hamburg, Germany

APR. 2025 – AUG. 2025

Received the **highest possible grade** in *Numerical Treatment of Ordinary Differential Equations, Probability Theory* and the *Modelling Camp* research project.

- **Numerical Analysis & Scientific Computing:** explicit and implicit time integrators for stiff and non-stiff ODEs; finite-difference and finite-volume methods; Python implementations; stability and convergence analysis.

Semester I — University of L'Aquila, Italy

SEPT. 2024 – FEB. 2025

Awarded grade **30/30 cum laude in both** *Applied Partial Differential Equations* and *Control Systems,* and **29/30** in *Mathematical Modelling of Continuum Media.*

- **Continuum Mechanics & PDE Modelling:** conservation laws and continuum balance equations; incompressible Navier–Stokes and Stokes equations; advection–diffusion transport equations; stress and linear elasticity.

BSc — Economics, Mathematics & Statistics
JUN. 2019 – MAY 2022

CHRIST (Deemed to be University), Bengaluru, India

CGPA 9.7/10; coursework in probability and stochastic processes, mathematical modelling, time-series forecasting and statistical computing in R and Python.

SELECTED COMPUTATIONAL RESEARCH PROJECTS

Dynamic Micro–Macroscopic Hybrid Model for Efficient Highway Traffic Simulation
JUN. 2025 – SEPT. 2025

Modelling Camp research project · Department of Mathematics · University of Hamburg

- **Implemented a multi-fidelity micro–macro traffic framework** based on Imai et al. (2024), coupling the macroscopic Cell Transmission Model with the vehicle-level Intelligent Driver Model through adaptive resolution switching.
- **Developed and verified a Godunov finite-volume solver for the LWR conservation law** against exact solutions, validating characteristic propagation, shock and rarefaction dynamics, and mass conservation.
- **Designed mass-preserving macro–micro interface rules** using vehicle reservoirs and residual tracking, enabling consistent state transfer across model boundaries; verified global conservation across free-flow, congested, and bottleneck scenarios.
- **Implemented adaptive fidelity selection** using density-triggered hysteresis, dwell-time constraints, and a local coefficient-of-variation trigger, activating microscopic resolution only in congested regions.

Hyperparameter Optimisation with Approximate Gradient (HOAG)
JAN. 2026 – FEB. 2026

Fundamentals of Machine Learning project · Université Côte d'Azur

- Analysed **bilevel hyperparameter optimisation via implicit differentiation**, including the $O(\varepsilon_k)$ hypergradient-error bound and convergence under summable inner-solver tolerances (Pedregosa, ICML 2016).
- **Implemented HOAG in Python** for ℓ_2-regularised logistic regression and RBF kernel ridge regression using L-BFGS-B and matrix-free conjugate gradients, respectively; evaluated the method on high-dimensional sparse datasets.
- **Benchmarked HOAG** against grid and random search, iterative differentiation and Bayesian optimisation, comparing suboptimality under a fixed computational budget.

INDUSTRY EXPERIENCE

Associate Analyst — Deloitte Consulting (U.S.–India Office)
AUG. 2022 – JUN. 2024

Bengaluru, India

- Developed and maintained automated end-to-end regression tests for ERP data workflows, covering data ingestion, transformation, and validation to improve software reliability and reproducibility.
- Led a six-member team through monthly regression-test cycles, streamlining test execution, defect tracking, and reporting to reduce overall cycle time.

RESEARCH & TECHNICAL SKILLS

- **Numerical Modelling & Scientific Computing:** operator splitting for hybrid continuous–discrete systems; CFL-based timestep control; k-d-tree candidate searches, spatial sectoring and binary-heap event scheduling; timestep-refinement and conservation diagnostics.
- **Carrier-Field Reconstruction & Flow Diagnostics:** multidimensional space–time interpolation using periodic cubic B-splines, quadratic Lagrange interpolation and Catmull–Rom/Hermite splines; FFT-based spectral differentiation; velocity-gradient, strain, vorticity, divergence and spatial-autocorrelation diagnostics.
- **Reduced Continuum & Stochastic Modelling:** particle-response-time expansions; stochastic closure of unresolved collision-history terms through collision-localised random forcing; reduced particle-density evolution equations.
- **Transport Statistics & Uncertainty Quantification:** radial distribution functions, correlation dimensions, DBSCAN morphology, pair/event statistics, moving-block bootstrap, confidence bands, and sensitivity analysis.
- **Optimisation & Machine Learning:** bilevel optimisation and implicit differentiation; L-BFGS-B; neural-network training in PyTorch using SGD and Adam.
- **Programming & Tools**: Python (NumPy, PyTorch, SciPy, Numba, scikit-learn, pandas, Matplotlib); R; Git; Linux; LaTeX/TikZ.

AWARDS & SCHOLARSHIPS

Machine Learning + Simulation Thematic Semester — Inria (2026) · Selected and funded for master's thesis research with the CALISTO team at Inria Sophia Antipolis.

MAECI Scholarship (2024–25) — Italian Ministry of Foreign Affairs · Merit-based funding for tuition and living costs.

University Merit Scholarship (2020–21 & 2021–22) — awarded for outstanding academic performance at CHRIST (Deemed to be University).